Examining the Association Between Laboratory Biomarkers and Bipolar Disorder Diagnosis: A Data-Driven Analysis

Selenge Bat-Erdene
Dalla Lana School of Public Health, University of Toronto
Toronto, Ontario, Canada
selenge.erdene@mail.utoronto.ca

Uththami Kukathasan
Dalla Lana School of Public Health, University of Toronto
Toronto, Ontario, Canada
uththami.kukathasan@mail.utoronto.ca

Victoria Truong
Dalla Lana School of Public Health, University of Toronto
Toronto, Ontario, Canada
victoria.truong@mail.utoronto.ca

Erwen Wu
Dalla Lana School of Public Health, University of Toronto
Toronto, Ontario, Canada
erwen.wu@mail.utoronto.ca

Abstract
Bipolar disorder (BD) is a chronic psychiatric condition characterized by recurrent mood episodes, often misdiagnosed due to symptomatic overlap with other mood disorders, leading to delayed treatment and adverse outcomes. Emerging evidence suggests metabolic dysregulation, particularly in glucose and lipid profiles, may contribute to BD pathophysiology, but findings on their diagnostic utility remain inconsistent. Using data from the Canadian Primary Care Sentinel Surveillance Network (CPCSSN), we analyzed 154 patients (318 observations) to investigate whether fasting plasma glucose (FPG), total cholesterol (TC), low-density lipoprotein (LDL), and high-density lipoprotein (HDL) can distinguish BD subtypes (manic vs unspecified). Through exploratory data analysis, statistical tests, K-means clustering, and logistic regression, we found total cholesterol (p=0.018) and sex (p=0.014) were significantly associated with BD subtypes, with higher cholesterol linked to lower odds of mania. However, the model explained only 5.43% of outcome variability, aligning with conflicting literature on biomarker specificity. These findings indicate that FPG, TC, LDL, and HDL may have limited utility as standalone diagnostic markers for BD subtype differentiation. Future studies should incorporate larger, longitudinal datasets and integrate inflammatory, hormonal, genetic, microbiome, and psychological factors using advanced machine learning approaches to improve subtype differentiation in bipolar disorder.

CCS Concepts
- Applied Computing → Health Informatics
- Information Systems → Data Cleaning

Keywords
Bipolar disorder, bipolar disorder subtypes, laboratory biomarkers, glucose markers, lipid markers, diagnostic accuracy, electronic health records, data wrangling, correlation analysis.

1. Introduction
Bipolar disorder (BD) is a chronic, recurrent psychiatric condition characterized by extreme mood fluctuations, ranging from manic or hypomanic episodes to severe depression, affecting approximately 2–2.5% of the global population [1,2]. The disorder imposes a significant burden, with an estimated economic cost of US$151 billion in 2009, driven by frequent relapses, hospitalizations, and impaired social and occupational functioning [1]. BD is associated with substantial psychiatric and medical comorbidities, including anxiety disorders (65–90% lifetime prevalence), substance use disorders (up to 60%), and metabolic conditions such as type 2 diabetes and cardiovascular disease, which contribute to a reduced life expectancy of approximately 10 years [2,3]. These comorbidities, coupled with high suicide rates, underscore the urgent need for effective diagnostic and management strategies [2].

Accurate diagnosis of BD remains a significant challenge, with misdiagnosis rates as high as 40%, often as unipolar depression, due to overlapping symptoms and the reliance on subjective clinical assessments [2]. Diagnostic delays, averaging 5–10 years, exacerbate clinical outcomes by increasing episode frequency, worsening functional impairment, and elevating the risk of comorbidities [1,2]. The absence of validated biomarkers further complicates differentiation of BD from other mood disorders, as current diagnostic criteria (e.g., DSM-5, ICD-10) depend heavily on phenomenology-based evaluations, which are prone to subjective interpretation [4].

Bipolar disorder and related conditions are classified in the ICD-9-CM under diagnostic codes 296.0 to 296.89. These codes reflect different clinical presentations: for example, a single manic episode (296.0), recurrent manic episodes (296.1), or a current manic (296.4), depressive (296.5), mixed (296.6), or unspecified episode (296.7). Broader categories include unspecified bipolar disorder (296.80) and other specified forms (296.89). These coding distinctions align with the clinical classification of bipolar disorder into Bipolar I disorder (requiring at least one manic episode), Bipolar II disorder (involving hypomanic and major depressive episodes), cyclothymic disorder (chronic, subthreshold mood fluctuations), and other specified or unspecified forms that do not meet full criteria. This diagnostic framework helps differentiate bipolar disorder from unipolar depression. However, it depends largely on subjective clinical evaluation, which limits accuracy and consistency [2]. As a result, misdiagnosis is common and may lead to inappropriate treatment (for example, prescribing antidepressants without mood stabilizers) which can provoke manic episodes or rapid cycling, complicating the course of illness and patient outcomes [1,3].

Emerging evidence suggests that BD's complex pathophysiology, involving neuroinflammation, oxidative stress, and mitochondrial dysfunction, may manifest in systemic metabolic changes, including abnormalities in glucose and lipid profiles [4,5]. These metabolic alterations are linked to the high prevalence of metabolic syndrome and cardiovascular comorbidities in BD, which are exacerbated by psychotropic medications such as atypical antipsychotics and mood stabilizers [1,3]. Peripheral biomarkers, such as fasting plasma glucose (FPG), glycated hemoglobin (HbA1c), triglycerides (TGA), total cholesterol (TC), low-density lipoprotein (LDL), and high-density lipoprotein (HDL), are accessible, cost-effective, and routinely collected in primary care settings, making them promising candidates for diagnostic indicators [4]. Staging models of BD suggest that biomarker abnormalities may be more pronounced in early stages, offering potential for early detection and intervention to prevent disease progression [6].

1.1 Laboratory Indicators of Bipolar Disorder

While BD is typically diagnosed through clinical assessments, recent research has examined biochemical markers from blood and urine samples as potential indicators or predictors of the disorder [7].

1.1.1 History of Markers in Mental Health Diagnoses

The use of blood to evaluate mental illness dates to the mid-19th century, when clinicians sought to distinguish sick patients from the healthy individuals through blood analysis [8]. A pivotal advance came in 1954, when Henneman et al. examined glucose metabolism in patients with schizophrenia, manic-depressive illness (now bipolar disorder), and involutional psychoses, identifying pronounced metabolic abnormalities in schizophrenia, independent of medication effects [9]. Their findings catalyzed decades of research into the interplay between metabolism and psychiatric disorders. In bipolar disorder research, ongoing studies continue to assess the diagnostic relevance of markers such as fasting glucose, glucose tolerance, HbA1c, triglycerides, total cholesterol, LDL, and HDL.

1.1.2 Glucose Biomarkers: FPG and HbA1c

Abnormal glucose regulation has emerged as a key area of interest for understanding the physiological basis of BD. Several studies have investigated markers such as FPG, glucose tolerance, and HbA1c to evaluate their relevance in BD diagnosis and progression. Garcia-Rizo et al. found that 71% of first-episode BD patients exhibited abnormalities on oral glucose tolerance tests, indicating glucose dysregulation early on in patients with the disease [10]. Age-related trends have also been observed, with FPG levels peaking among BD patients aged 40-60, which could be a result of the increased likelihood of type-2 diabetes among individuals aged 45 and older [11]. When comparing BD to other mood disorders, Tan et al. reported that BD patients had significantly lower FPG levels, suggesting that elevated glucose might be more indicative of non-BD conditions [7].

In addition to short-term glucose measures, HbA1c, which is a longer-term marker of glucose metabolism, has also been explored in BD populations. While some studies report lower HbA1c levels in BD patients relative to those with other mood disorders, others have found substantial variation within the BD population [7]. A study by Castilla-Puentes et al. observed that over half of 77,000 BD patients had elevated HbA1c levels, indicating metabolic vulnerability to impaired glucose metabolism [12]. Additionally, Fellendorf found that among men with BD, higher HbA1c levels were associated with a greater number of depressive and manic episodes, suggesting a link between glycemic control and illness severity [13].

Despite the potential value of glucose markers, their clinical use may be limited by under-testing in psychiatric populations. A study by Bell et al. examined electronic health records from 2008-2018 and found that only 19.1% of patients with mood disorders had at least one HbA1c lab record [14]. The study also found disparities in testing rates, with African and non-Caucasian groups having more HbA1c results than Caucasian groups, and men were more likely to be tested than women. Younger patients aged 20-29 had the highest frequency of HbA1c records, with testing rates declining with age. Therefore, one of the main challenges in translating biomarker research into clinical practice is the lack of routine testing for key lab markers like HbA1c in patients with mood disorders. This limits the ability to identify associations between lab markers and conditions such as BD.

1.1.3 Lipid Biomarkers: TGA, TC, LDL, and HDL

Lipid abnormalities involving TGA, TC, LDL, and HDL have been studied as potential biomarkers for BD [10]. Wysokiński et al. found that BD patients had an increased risk of elevated TGA levels, often surpassing the upper normal limit of 150 mg/dL, with levels varying significantly across age groups [15]. Similarly, TC was also elevated in BD patients compared to those with other mood disorders, with a greater risk of TC levels exceeding 200 mg/dL and significant age-related trends [15].

In contrast to TGA and TC, LDL levels appeared to be lower in individuals with BD. Tan et al. reported a negative correlation between LDL levels and BD, where lower LDL concentrations were more frequently observed in patients with BD [7]. This finding suggests that LDL may serve as an informative marker in differentiating BD from other psychiatric conditions [7]. However, not all lipid markers show consistent associations; Tan et al. also found no significant differences in HDL, TC, or TGA levels across groups, raising questions about the reliability of some lipid measures as diagnostic indicators.

Differences in lipid profiles may also reflect underlying distinctions between BD subtypes and states. A study by Fusar-Poli et al. observed that patients with manic episodes had significantly lower TC, HDL, and LDL levels compared to euthymic BD, which is a subtype under the unspecified subtype for the ICD-9 codes [16]. Additionally, manic patients also had significantly lower TC and LDL than those with the depressive subtype. As for TGA levels, researchers found no significant differences across the three subtypes.

These findings are supported by Huang et al., who demonstrated that lipid profiles are not only subtype-dependent but also state-dependent [17]. In their study of patients with bipolar I disorder, TC and TGA levels were significantly lower during acute manic episodes compared to depressive states. Researchers also found that the lowest rate of dyslipidemia, which is elevated TGA or reduced HDL, was observed during acute mania. This suggests that lipid abnormalities in BD may fluctuate with mood polarity and symptom severity, reinforcing the potential of lipid markers as dynamic indicators of illness state.

While lipid markers have received increasing attention, other biomarkers such as coagulation markers remain underexplored as the literature surrounding INR in BD is sparse [10]. Garcia-Rizo noted a lack of studies examining the relationship between INR and BD, highlighting a significant research gap. They may be partly because INR is not routinely measured during standard blood tests, as it is primarily used to monitor anticoagulation therapy [18]. Other markers like platelet count or prothrombin time may offer broader insight into inflammation and coagulation processes in BD [19]. Despite these limitations, such markers require the need for further exploration to assess their potential role in differentiating BD from other mood disorders.

1.2 Motivation for Proposed Research

The pressing need for objective diagnostic tools in bipolar disorder (BD) motivates this research, as high misdiagnosis rates and delays lead to inappropriate treatments and worsened outcomes, driven by reliance on subjective criteria [3]. BD's pathophysiology, involving neuroinflammation and oxidative stress, manifests in metabolic dysregulation, yet inconsistent lipid (triglycerides, total cholesterol, HDL) and glucose (fasting plasma glucose, HbA1c) marker findings highlight gaps in reliable biomarkers [3]. This study leverages CPCSSN data to validate these accessible biomarkers, aiming to improve diagnostic accuracy and guide early interventions in primary care.

2. Research Question

Do fasting plasma glucose, HbA1c, total cholesterol, low-density lipoprotein, and high-density lipoprotein, as routinely collected biomarkers, differ significantly between bipolar disorder subtypes in primary care patients from the CPCSSN database, and can these biomarkers, analyzed through Welch's t-tests, Chi-square tests, K-means clustering, and logistic regression, enhance diagnostic accuracy when combined with clinical assessments?

3. Data and Variables

The CPCSSN provided 11 data tables comprising over 20 million patient records collected across a span of more than 10 years. Text diagnoses made by physicians were mapped to the International Classification of Diseases, 9th Revision (ICD-9) codes, and then stored as data tables. For this analysis, three primary data tables are used: `C4MLab.csv, C4MEncounterDiagnosis.csv, and C4MPatient.csv.` Additional data tables are used for validation purposes.

3.1 C4MLab.csv

This table contains laboratory test results, including the biomarkers of interest for this study. Relevant measures include lipid markers (Total Cholesterol, LDL, HDL), glucose markers (Fasting Glucose, HbA1c), and coagulation markers such as INR. Relevant variables in this data table include `Patient_ID`, `TestResult_calc` (the result of the lab test recoded into consistent text), and `PerformedDate` (date the lab test was done).

3.2 C4MEncounterDiagnosis.csv

This dataset captures all diagnoses recorded during an encounter with a patient. In this particular study, the relevant diagnosis would be BD, identified by an ICD-9 code of 296. Relevant variables in this data table include `Patient_ID`, `DiagnosisCode_calc` (diagnosis code recoded into consistent text), and `DateCreated` (date stamp of the record).

3.3 C4MPatient.csv

This data table contains important demographic information such as sex and birth year (which can be used to determine age). These demographic variables can be used as auxiliary variables in the imputation and modeling processes. Relevant variables in this data table include `Patient_ID`, `Sex`, and `BirthYear`.

3.4 C4MHealthCondition.csv

This data table contains information regarding other health conditions and comorbidities that a patient may have at the same time as BD. It contains important variables such as `Patient_ID`, `DiagnosisCode_calc`, and `DateCreated` that can help during the matching process, and it can help verify health conditions and mental health diagnoses.

4. Methods

The methodological framework, depicted in Figure 1, outlines a structured pipeline comprising cohort selection based on initial BD diagnosis criteria, data wrangling and preprocessing to standardize lab results and demographic variables (age, sex, comorbidity), and k-nearest neighbours (KNN) imputation for missing data. Subsequent analyses, including exploratory data analysis, Welch's T-tests, Chi-square tests, K-means clustering and logistic regression, are applied to evaluate these biomarkers' diagnostic potential and their correlations with clinical and demographic factors in a primary care context.

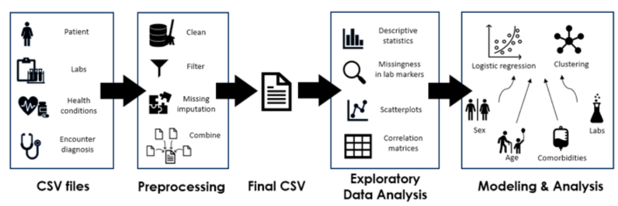

Figure 1. *Schematic diagram of project pipeline*

4.1 Cohort Selection Criteria

To identify and select our study population, strict diagnosis-based inclusion criteria were applied. We identified patients with BD using eight ICD-9 codes and ensured that their BD diagnosis was their first diagnosis ever recorded. This date was used as the index date for all subsequent analyses, thereby restricting all lab results and comorbid conditions included in the study to those occurring after the initial BD diagnosis. This approach reduced potential bias from pre-existing conditions.

To evaluate lab data availability, we assessed whether each patient had at least one lab result for any of the seven selected lab markers. Patients were categorized based on the timing of the lab results relative to their BD diagnosis: before, after, or both.

We also examined whether patients had other diagnoses recorded on the same day as their BD diagnosis. To prevent misclassification of routine visits or screenings as comorbid conditions, we excluded diagnoses with ICD-9 codes that began with `"V"` (e.g., medical exams, counseling) as well as records with `"nan"` strings. All ICD-9 codes were standardized by uppercasing and removing whitespace to ensure consistency.

Following this initial exploration, we observed that the majority of BD patients had labs recorded after their diagnosis. To focus on the temporally relevant, post-diagnosis cohort, we restricted the final sample to patients who had at least one lab test within two years (730 days) following their BD diagnosis. This time window was chosen to balance clinical relevance with sufficient sample size. Each patient was treated as a unique observation in subsequent analyses.

4.2 Data Wrangling and Preprocessing

For patients meeting the final inclusion criteria, lab test data were first converted from long to wide format, with each row representing a lab encounter and each column representing a specific lab marker. Patient-level information such as sex, age at the time of testing, BD diagnosis date, and BD diagnosis code were merged into the dataset.

If a patient had multiple lab entries on the same day, these were combined into a single row containing the most complete set of available lab results. The data were then re-pivoted to ensure consistency across lab encounters.

Since it is common for patients to develop comorbidities in the period following a BD diagnosis, we created a binary Comorbidity variable. Patients were assigned a value of 1 if they received any non-BD diagnosis on the same day as, or within two years of their BD diagnosis. Those with only routine medical visits or no additional diagnoses were assigned a value of 0.

For patients with multiple lab entries, the Comorbidity label could vary over time. For example, a patient whose lab result occurred shortly after their BD diagnosis would initially receive a `0`. However, if that same patient was later diagnosed with a condition such as diabetes during the two-year window, all subsequent lab records after that diagnosis would be flagged with a `1`. This approach helps identify patients with potentially more complex clinical profiles whose lab values may differ from those without comorbid conditions.

4.3 Missing Data Imputation

After loading the pre-processed dataset, observations with missing values for age and sex were removed, to ensure the accuracy of future imputation and modelling processes. To prepare the sex variable for imputation and modelling, all values were standardized to consistent case formatting, and numerical values were assigned (0 for female, 1 for male).

Lab markers with more than 60% missing data were also removed to avoid introducing unnecessary uncertainty during the imputation process. The 60% threshold was chosen to balance retaining as many lab variables as possible, while also ensuring data quality.

To enhance the predictive accuracy of the imputation process, age and sex were temporarily scaled by a factor of two. This weighting was applied so that these demographic characteristics would have greater influence in determining the similarity between records during the imputation procedure. Since age and sex are important determinants of many laboratory measures, increasing their influence in the similarity calculations helped ensure that imputed values came from patients with comparable demographic profiles. This approach reduced the risk of imputing values based on variables that are less biologically relevant.

Missing values for the remaining laboratory markers were imputed using k-nearest neighbours (KNN) imputation. This method identifies the k most similar records (here, $k = 5$) based on a distance metric and estimates missing values using information from those eighbours. A uniform weighting scheme was used, meaning all 5 neighbours contribute equally and carry the same weight. KNN imputation was chosen because the dataset is very sparse, with most patients only having 2-3 lab markers recorded. KNN is a non-parametric method that can use partial information from similar patients to impute missing values, instead of requiring complete cases. Age and sex were included as predictors in the imputation but were excluded from being directly imputed. After imputation, age and sex were rescaled back to their original values for interpretability in future modelling. The resulting cleaned and imputed dataset was saved as a csv file for further analysis.

4.4 Exploratory Data Analysis

Exploratory data analysis (EDA) was conducted to characterize the dataset, evaluate the distributions of key variables (fasting plasma glucose [FPG], high-density lipoprotein [HDL], low-density lipoprotein [LDL], total cholesterol [TC], age, sex, and comorbidity), examine inter-variable relationships, and validate assumptions for logistic regression and K-means clustering models. The analysis was performed using Python (version 3.8) with standard statistical and visualisation libraries to assess data patterns and inform subsequent modelling.

The dataset's structure was evaluated to confirm its dimensions, variable types, and completeness following imputation. Descriptive statistics, including mean, median, standard deviation, skewness, kurtosis, percentiles, and interquartile range, were calculated for continuous variables (FPG, HDL, LDL, TC, age) to summarize their distributional properties and identify deviations from normality.

Histograms with kernel density estimation and normal distribution overlays were generated to assess the distributions of FPG, HDL, LDL, TC, and age, both overall and stratified by bipolar disorder (BD) subtype (manic vs. unspecified). These visualizations aimed to evaluate normality and skewness, informing assumptions for statistical tests used in logistic regression.

Bivariate relationships were explored through scatter plots of variable pairs (FPG vs. HDL, LDL vs. TC, age vs. FPG), stratified by BD subtype, to assess linearity and subtype differentiation. Pearson and Spearman correlation analyses quantified associations among continuous variables and checked for multicollinearity. Group comparisons were performed to compute means and standard deviations of biomarkers by sex, comorbidity, and BD subtype. Cross-tabulations examined associations between categorical variables (sex, comorbidity) and BD subtype.

To verify the assumption of equal variances for group comparison tests, Levene's test was applied to FPG, HDL, LDL, TC, and age between manic and unspecified groups. Test statistics and p-values were calculated, with $p < 0.05$ indicating unequal variances, necessitating adjusted statistical tests (e.g., Welch's t-test).

Violin plots were used to compare biomarker distributions across BD subtype, sex, and comorbidity. Lollipop charts visualised mean differences by BD subtype, and percentile plots explored biomarker trends across age groups to identify age-related patterns.

4.5 Modelling outcomes and predictors

The outcome of interest is bipolar disorder status, recorded in the variable State_Type, where manic episodes are coded as 1 and unspecified episodes as 0. Specifically, the manic group includes diagnostic codes 296.0, 296.1, 296.4, and 296.6, while the unspecified group includes codes 296.7 and 296.8. The depressed group initially included code 296.5. However, when we added the condition that the same patient must also be diagnosed with a condition such as diabetes within the two-year window, we no longer had any observations for depressed patients. As a result, the imputed dataset only contains two groups: Manic and Unspecified. The predictor variables include FPG, HDL, LDL, TC, age, sex, and comorbidity. Both sex and comorbidity are binary variables, coded as 0 and 1. When modeling the association, sex and comorbidity will be treated as categorical variables (factors).

4.6 Welch's T-test for Continuous Variables

We used a Welch's t-test to determine whether there is a statistically significant difference between the manic group and unspecified group for the continuous predictors which are FPG, HDL, LDL, TC, and age.
Assumptions:

- Manic group and unspecified group are independent.
- Variables should be continuous.
- Data follows a normal distribution. (verified in 5.3.2)
- The variances of the two groups are allowed to unequal. (verified in 5.3.3)

4.7 Chi-square test for Categorical Variables

We used a Chi-square test to determine whether there is a statistically significant difference between the manic group and unspecified group for the categorical variables including sex and comorbidity.
Assumptions:

- Independence of Observations.
- Frequency in each cell should be ≥ 5.

4.8 Logistic regression

Logistic regression, a type of generalized linear model designed for binary outcomes, was used in this analysis. The outcome variable, State_Type, was coded as Manic (1) versus Unspecified (0). This modeling approach allowed us to quantify the relationship between selected biomarkers and the likelihood of an individual being in the manic state. Our predictors are FPG, HDL, LDL, TC, age, sex, and comorbidities.
Our model is:

$$logit\left(\frac{P(y=1)}{P(y=0)}\right) = \beta_0 + \sum \beta_i x_{ij} + \epsilon$$

Where:
y is Status_Type where manic is 1 and unspecified is 0
x_{ij} is i^{th} predictor of observation j

4.9 K-means clustering

K-Means clustering is an unsupervised machine learning algorithm that partitions data into K distinct groups based on distance metrics. In our study, the goal of the algorithm is to organize data points into clusters where points in the same cluster share similar characteristics, and points in different clusters are different. The algorithm iteratively assigns each data point to the nearest cluster centroid and updates each centroid with the mean of the data points assigned to it. This process repeats until cluster assignments stabilize or changes in centroid positions fall below a predefined threshold.

The K-means algorithm tries to minimize the Within-Cluster Sum of Squares (WCSS):

$$\sum_i \sum_x ||x - \mu_i||^2$$

Where:
x is vector $(x_1, x_2, x_3, \dots, x_n)$
μ_i is the mean of points in S_i, $\mu_i = \frac{1}{S_i}\sum_x xS_i$
S_i is the set of points assigned to the i^{th} cluster

5. Results

A cohort of 154 patients (318 observations) with bipolar disorder (BD) was derived from the CPCSSN database, focusing on laboratory biomarkers (FPG, HDL, LDL, TC) and demographic variables (age, sex, comorbidity) within two years post-BD diagnosis. Missing data were imputed using k-nearest neighbours (*k=5*). Distributions, variances, and correlations were assessed through exploratory data analysis. Levene's test confirmed unequal variances for HDL, LDL, and age, necessitating Welch's T-test for group comparisons. Chi-square tests evaluated associations for categorical variables, ensuring expected cell frequencies ≥5. Logistic regression modeled biomarker and demographic associations with BD subtypes (manic vs unspecified). K-means clustering, with *k=2* selected via the elbow method, explored natural groupings. These analyses assessed the diagnostic potential of biomarkers and demographics in distinguishing BD subtypes, addressing their statistical significance and predictive utility.

5.1 Cohort Selection

From the 352,161 patients in the CPCSSN database, 378 individuals were identified as having BD as their first-ever recorded diagnosis. Of these, 218 patients had available lab results for at least one of the seven selected biomarkers. Among them, 4 patients had lab records prior to their BD diagnosis and were excluded due to the limited sample size in that subgroup. The remaining 214 patients had lab results occurring after their BD diagnosis.

To focus on the early post-diagnosis period, we further restricted the cohort to patients with lab records within two years of their BD diagnosis date. This resulted in a final cohort of 154 patients, excluding 60 patients whose lab tests occurred beyond the two-year window. Within this cohort, 124 patients had at least one comorbidity diagnosis during the two-year period, while 30 had no comorbidities. The pre-processed dataset contains 319 lab observations across 154 patients. Figure 2 summarizes the inclusion and exclusion criteria used to define the final study cohort.

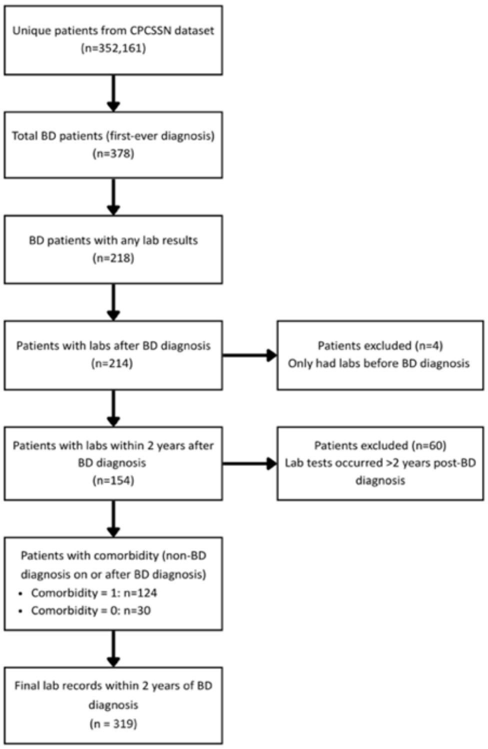

Figure 2. *Cohort selection flow diagram displaying the inclusion criteria and filtering steps for bipolar disorder (BD) patients from the Canadian Primary Care Sentinel Surveillance Network (CPCSSN) dataset.*

5.2 Missing Imputation of Lab Markers

The initial pre-processed dataset contained 319 observations, with one observation excluded due to missing age, resulting in 318 patients for analysis. Three laboratory markers (glucose tolerance, INR, HbA1c) were removed due to high proportions of missing data (>60%), leaving four retained markers: LDL, HDL, TC, and FPG, as shown in Table 1 below:

Table 1*: Missing Data for Each Lab Marker*

Lab Marker	Missing Data Percentage
Glucose Tolerance	99.69
INR	77.04
HbA1C	67.92
LDL	50.00
HDL	49.37
Total Cholesterol	45.28
Fasting Plasma Glucose	40.25

Among the retained markers, missingness ranged from 40.25% (fasting glucose) to 50.00% (LDL). In total, 588 missing values across these markers were estimated using KNN imputation (k=5, uniform weighting). To improve the quality of imputed estimates, age and sex were scaled by a factor of two prior to imputation and later rescaled back to their original values. Post-imputation, the dataset was fully complete for the retained laboratory markers.

5.3 Exploratory Data Analysis

5.3.1 Descriptive statistics

Descriptive statistics for continuous variables (n=318) after imputation were as follows (Table 2):

Table 2: *Descriptive statistics for continuous numeric variables*

Variable	Mean (SD)	Median	Skewness	Kurtosis	Range	IQR
FPG (mmol/L)	5.96 (2.03)	5.40	4.03	21.54	3.60 –21.50	1.24
HDL (mmol/L)	1.32 (0.31)	1.29	0.96	2.46	0.43 – 2.60	0.28
LDL (mmol/L)	2.93 (0.75)	2.96	0.30	1.16	0.83 – 5.60	0.73
TC (mmol/L)	4.87 (1.06)	4.93	-0.52	3.55	0.88 – 8.74	0.91
Age (years)	52.81 (16.75)	51.00	0.64	-0.26	21 – 88	20.00

The pronounced skewness in FPG (4.03) indicated a right-skewed distribution, potentially affecting statistical assumptions for parametric tests. HDL and age showed moderate skewness (0·96 and 0.64, respectively), while LDL and TC approximated normality (skewness 0.30 and –0.52).

5.3.2 Distribution Analysis

Histogram analyses revealed right-skewed distributions for FPG, moderate skewness for HDL, and near-normal distributions for LDL and TC (Figure 3). When stratified by BD subtype, distributions of FPG, HDL, LDL, and TC were similar between manic and unspecified groups, suggesting limited discriminative capacity (Figure 4 and 5, respectively).

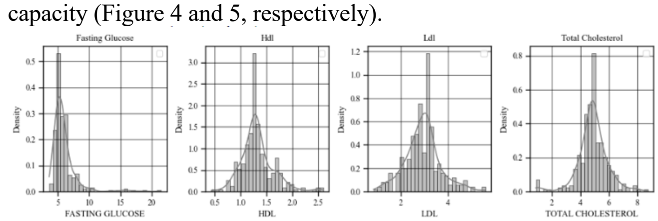

Figure 3. *Distributions of FPG, HDL, LDL, and Total Cholesterol*

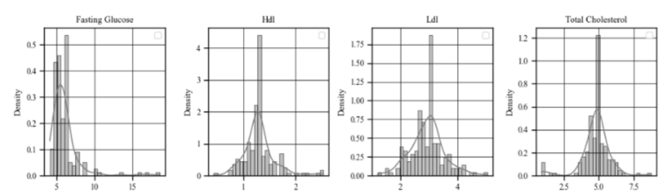

Figure 4. *Distributions of FPG, HDL, LDL, and Total Cholesterol - manic subset*

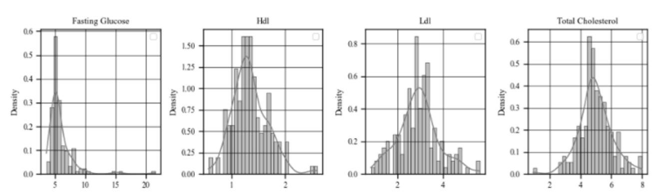

Figure 5.*Distributions of FPG, HDL, LDL, and Total Cholesterol-unspecified subset*

A violin plot (Figure 6) visualized age by subtype: the unspecified group showed a near-normal shape (mcedian 51.00 years, IQR 19.25), while the manic group displayed an hourglass shape (median 51.00 years, IQR 21.00), suggesting potential bimodal onset [10]. However, the overlap in the densities was noted as potentially indicative of limited diagnostic utility.

Figure 6. *Violin plot of age by unspecified (left) and manic (right) subtype*

5.3.3 Variance Equality

Levene's test assessed equality of variances between manic and unspecified subtypes:

- FPG: statistic 0.004, p=0.951 (equal variances).
- HDL: statistic 4.910, p=0.027 (unequal variances).
- LDL: statistic 13.817, p=0.0002 (unequal variances).
- TC: statistic 3.057, p=0.081 (equal variances).
- Age: statistic 14.217, p=0.0002 (unequal variances).

Unequal variances for HDL, LDL, and age necessitated the use of Welch's t-test for group comparisons.

5.3.4 Correlation Analysis

Correlation analyses assessed relationships among FPG, HDL, LDL, TC, and age. Pearson and Spearman correlations (Table 3) showed a moderate LDL-TC correlation given their biological relationship, below the multicollinearity threshold (>0.8). Weak correlations were observed for the remaining pairs.

Table 3: *Pearson and Spearman correlations*

Variable Pair	Pearson (p-value)	Spearman (p-value)
FPG–HDL	-0.121 (0.031)	-0.050 (0.374)
FPG–LDL	-0.200 (0.0003)	-0.123 (0.028)
FPG–TC	-0.008 (0.881)	0.008 (0.892)
FPG–Age	0.097 (0.084)	0.296 (<0.0001)
HDL–LDL	-0.007 (0.904)	-0.010 (0.862)
HDL–TC	0.118 (0.035)	0.129 (0.022)
HDL–Age	0.015 (0.791)	0.143 (0.011)
LDL–TC	0.571 (<0.0001)	0.616 (<0.0001)
LDL–Age	0.077 (0.171)	0.174 (0.002)
TC–Age	-0.031 (0.585)	0.0003 (0.996)

5.3.5 Group Comparisons

Figure 7 presents box plots of FPG, HDL, LDL, TC, and age, stratified by BD subtype (manic, n=162; unspecified, n=156). Each plot displays the median (central line), IQR (box), whiskers (range excluding outliers), and individual data points.

Median FPG values are similar between manic (5.60 mmol/L) and unspecified (5.30 mmol/L) subtypes, with comparable IQRs (1.18 vs. 1.10 mmol/L) and few outliers. HDL (manic: 1.29 mmol/L; unspecified: 1.30 mmol/L), LDL (manic: 3.05 mmol/L; unspecified: 2.87 mmol/L), and TC (manic: 4.93 mmol/L; unspecified: 4.81 mmol/L) show overlapping medians and similar IQRs, indicating no distinct metabolic differences. Age shows a similar median (51.0 years for both), but greater variability in the manic subtype (IQR: 21.0 vs. 19.3 years), aligning with a significant age difference (p=0.0465, Table 3). These results suggest limited biomarker differentiation between BD subtypes, consistent with prior studies [4,7].

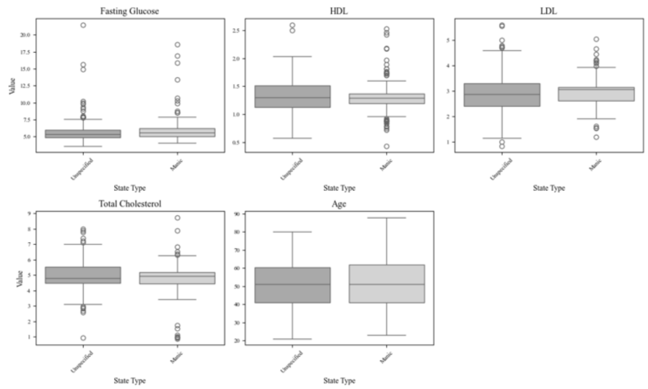

Figure 7*: Box Plots of FPG, HDL, LDL, TC, and Age (left to right) by BD Subtype*

Among 318 patients with bipolar disorder, 96.5% (n=307) had comorbidities, as shown in a cross-tabulation heatmap by comorbidity and state type. Patients with comorbidities were older (median 51 vs. 42 years, p=0.002) and had lower total cholesterol (median 4.93 vs. 5.25 mmol/L, p=0.046) than those without (n=11). Fasting glucose showed similar medians (5.4 mmol/L) but higher skewness with comorbidities (4.05 vs. 0.49). As seen from the cross-tabulation heatmap by comorbidity and state type, due to the imbalance in group sizes, these differences require cautious interpretation.

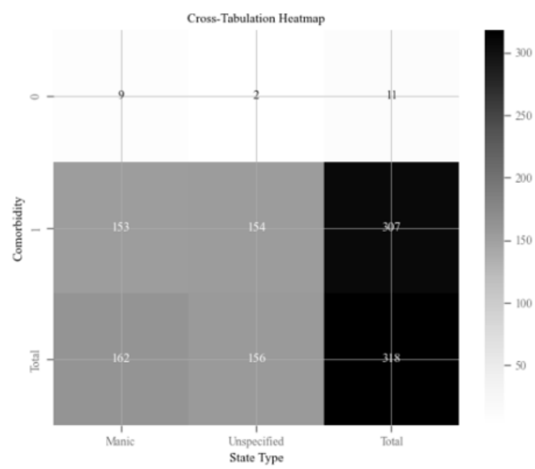

Figure 8*: Cross-Tabulation Heatmap by Comorbidity and State Type*

5.4 Modelling

Continuous variables (FPG, HDL, LDL, TC, age) were compared between manic and unspecified bipolar disorder subtypes using Welch's T-test. Associations between categorical variables (sex, comorbidity) and subtypes were evaluated with Chi-square tests, with contingency tables verified to have expected cell frequencies ≥5. The relationship between laboratory markers, demographics, and subtype classification was modeled using logistic regression. Natural groupings based on predictor variables were identified through K-means clustering, with the optimal number of clusters (k=2) determined by the elbow method. The utility of biomarkers and demographics in distinguishing bipolar disorder subtypes was assessed through these analyses.

5.4.1 Welch's T-test & Chi-square test

In our study, we included two categorical variables: sex and comorbidity. To meet the assumptions of the Chi-square test, which require that the expected frequency in each cell of the contingency table be at least 5, we examined the contingency tables for sex and comorbidity separately.

Table 4: *Contingency Table for Categorical Variables*

Sex	Manic	Unspecified
0	72	95
1	90	61
Comorbidity	**Manic**	**Unspecified**
0	9	2
1	153	154

Table 4 presents the contingency tables for these categorical variables. As shown, all cells have expected counts greater than or equal to 5, thereby satisfying the assumptions of the Chi-square test.

We then used a Welch's T-test to examine statistically significant differences in continuous variables and applied the Chi-square test for categorical variables.

Table 5: *T-test & Chi-square test results*

Variable	Test	Statistic	p-value
Fasting Plasma Glucose	Welch's t-test	0.834	0.405
HDL	Welch's t-test	-1.189	0.235
LDL	Welch's t-test	0.232	0.817
Total Cholesterol	Welch's t-test	-1.673	0.095
Age	Welch's t-test	2.012	0.045
Sex	Chi-square	7.980	0.005
Comorbidity	Chi-square	3.161	0.075

Table 5 presents the results of the Welch's T-test and Chi-square test. They assess statistically significant differences between the two bipolar disorder groups for each variable. Among the continuous variables, age was significantly different between the manic and unspecified groups (*p=0.045*) at the 5% significance level. The other variables, including FPG, HDL, LDL, and TC, did not show statistically significant differences. For the categorical variables, sex showed a significant association with group (*p=0.005*), while comorbidity did not (*p=0.075*).

5.4.2 Logistic regression

We used logistic regression to examine the association between laboratory markers and bipolar disorder. The estimated model coefficients are presented in Table 6.

Table 6: *Logistic Regression Estimates*

	Coef.	Std.Err	z	P>|z|
const	1.103	1.268	0.869	0.385
Fasting Plasma Glucose	0.086	0.062	1.383	0.167
HDL	-0.042	0.401	-0.104	0.917
LDL	0.313	0.214	1.461	0.144
Total Cholesterol	-0.368	0.155	-2.376	0.018
Age	0.012	0.007	1.721	0.085
Sex	0.622	0.254	2.446	0.014
Comorbidity	-1.637	0.837	-1.956	0.051

The logistic regression model converged successfully, and the likelihood ratio test indicated that the model fits significantly better than the null (intercept-only) model (*LLR p-value = 0.0012 < 0.05*).

Total cholesterol was significantly associated with the outcome (*β = -0.368, p=0.018*), suggesting that higher total cholesterol levels are associated with lower odds of being in the Manic group compared to the Unspecified group. Sex was also significant (*β = 0.622, p=0.014*), indicating that males had higher odds of being classified as Manic than females. Other predictors are not statistically significant at the 5% significant level. Comorbidity, TC, and HDL were negatively associated with odds of manic BD while age, sex, FPG and LDL were positively associated with odds of manic BD.

However, the Pseudo R-squared was 0.0543, suggesting that the model explains only 5.43% of the variability in the outcome. We aimed to understand why the logistic regression model performed poorly. One possibility is that the logistic model may not be appropriate for this dataset. Alternatively, the lab markers themselves may not carry sufficient predictive information for distinguishing bipolar disorder subtypes.

Scatterplots and the correlation matrix in the EDA session indicated no clear signs of multicollinearity or non-linear relationships among predictors. These findings indirectly suggest that laboratory markers and basic patient characteristics may not effectively predict whether a patient is in the manic or unspecified subtype of bipolar disorder.

5.4.3 K-means Clustering

We applied an unsupervised machine learning approach to cluster the dataset using laboratory markers and patient demographic information. K-means clustering was used to identify natural groupings within the data based on the predictor variables. Based on the Elbow Method shown in Figure 9, the optimal number of clusters is *k=3*, as this is where the within-cluster sum of squares (WSS) shows a noticeable inflection point before leveling off. However, *k=2* provides the best balance between model complexity and within-cluster variability for few reasons. For example, two clusters are easier to interpret and visualize and a third cluster adds minimal clinical insight in our study. Lastly, in cases with a limited sample size, selecting fewer clusters can help prevent overfitting and ensure more stable results.

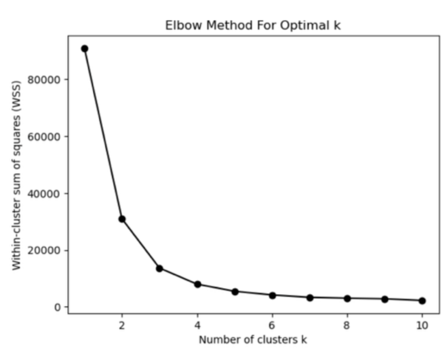

Figure 9*: Elbow Method for Optimal K*

Table 7: *K-means confusion matrix*

Cluster	Manic	Unspecified	Total
K-means Group 1	119	117	236
K-means Group 2	43	39	82

Table 7 presents the confusion matrix comparing the resulting K-means clusters with the defined manic and unspecified bipolar disorder groups. To be specific, manic and unspecified are the true clinical labels while K-means Group 1 & 2 are two clusters identified by K-means, based only on laboratory markers and demographic features. The clusters show poor alignment with the clinical subtypes. For instance, Group 1 contains 119 patients diagnosed as manic and 117 as unspecified, indicating a nearly even distribution. Similarly, Group 2 includes 43 manic and 39 unspecified cases, suggesting that the clustering did not generate a clear separation between the clinical subtypes.

We used a Chi-square test to assess the statistical association between the K-means clusters and bipolar disorder subtypes (manic & unspecified). The Chi-square statistic was 0.035 (*df=1, p=0.852*), indicating insufficient evidence to support an association between the clusters and bipolar disorder subtypes.

Table 8: *K-means Clusters Median Values*

Cluster	Fasting Plasma Glucose	HDL	Total Cholesterol	LDL
Group 1	5.30	1.25	4.86	2.86
Group 2	6.18	1.29	4.93	3.10
Cluster	**Age**	**Sex**	**Comorbidity**	
Group 1	46	0	1	
Group 2	76	0	1	

Table 8 presents the median values for each K-means cluster, highlighting both differences and similarities. FPG and age show the most notable distinctions: Group 2 has a higher median FPG level (6.18) compared to Group 1 (5.30). Additionally, Group 2 is obviously older than Group 1 (median age 76 vs. 46), which shows that age is likely the strongest driver of cluster separation. In contrast, the lab markers HDL, LDL, and TC are quite similar between clusters, indicating limited contribution to the clustering. Furthermore, the medians for sex and comorbidity in both clusters are identical, which means based on these two variables, there will not be meaningful separation. In conclusion, these results provide evidence that laboratory markers and selected demographic variables represent weak signals, insufficient for an unsupervised algorithm like K-means to identify meaningful. Furthermore, the findings suggest that important information may be missing from the dataset. Laboratory markers and basic demographic characteristics alone may not sufficiently capture the biological complexity that distinguishes bipolar disorder subtypes.

6. Limitations

This study has several limitations that may have influenced our findings. First, the sample size was relatively small, with 154 unique patients (318 total observations). Additionally, a large proportion of patients (309 out of 318) developed comorbidities within two years of their BD diagnosis, and 38 patients had non-BD diagnoses recorded on the same day. This overlap may have confounded the relationship between laboratory markers and BD subtypes. Our analysis was also limited to a few routinely collected biomarkers such as FPG, TC, LDL, and HDL, along with basic demographic factors. Important markers like HbA1c and INR were excluded due to high levels of missing data, which further narrowed the scope of our evaluation. Finally, both the logistic regression and K-means clustering analyses showed poor predictive performance, suggesting that these variables alone are not sufficient for distinguishing BD subtypes. Future studies should consider incorporating additional clinical, genetic, environmental, and psychosocial factors to better capture the complexity of BD.

7. Conclusions

This study explored whether common laboratory biomarkers and demographic characteristics could help distinguish between bipolar disorder subtypes. Similar to previous research reporting mixed and inconsistent findings [4, 7], our results suggest that these biomarkers alone are not reliable indicators of BD subtypes. While TC and sex appeared to be significant predictors in the logistic regression model, the model only explained 5.4% of the variance, and other markers were not statistically significant. Similarly, K-means clustering failed to identify meaningful groupings, highlighting the limitations of using unsupervised methods with these variables.

These findings reinforce earlier studies showing that metabolic dysregulation in BD, such as alterations in glucose and lipid profiles [10, 15] may reflect general metabolic dysregulation rather than subtype-specific patterns. This suggests that while these biomarkers are important for monitoring overall health in BD populations, they are unlikely to be used as strong diagnostic tools. Future research should focus on integrating different data sources, including genetic, neuroimaging, and detailed clinical information, to improve early detection and subtype classification.

8. Team Project Management

To manage our group project, we primarily used Discord to coordinate tasks, share updates, and resolve questions in real time. Regular team meetings, especially around key milestones, helped us stay aligned and track our progress. We divided responsibilities based on each member's strengths and used a shared Excel tracker to monitor tasks, deadlines, and individual contributions. In addition, GitHub allowed us to version-control our code and ensure everyone had access to the latest scripts and notebooks. This structured but flexible workflow helped us stay organized, meet our internal deadlines, and deliver a cohesive final project.

9. Acknowledgements

The use of large language models (e.g., ChatGPT) was used for grammar and to plan the formatting of the paper.

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